

Week3 #MONGODB

Thanam

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Step 1 — Open book_metadata

In MongoDB Atlas:

Database → Browse Collections → bookflow → book_metadata

You should see your existing documents.

Step 2 — Click Add Data

At the top of the book_metadata collection, click:

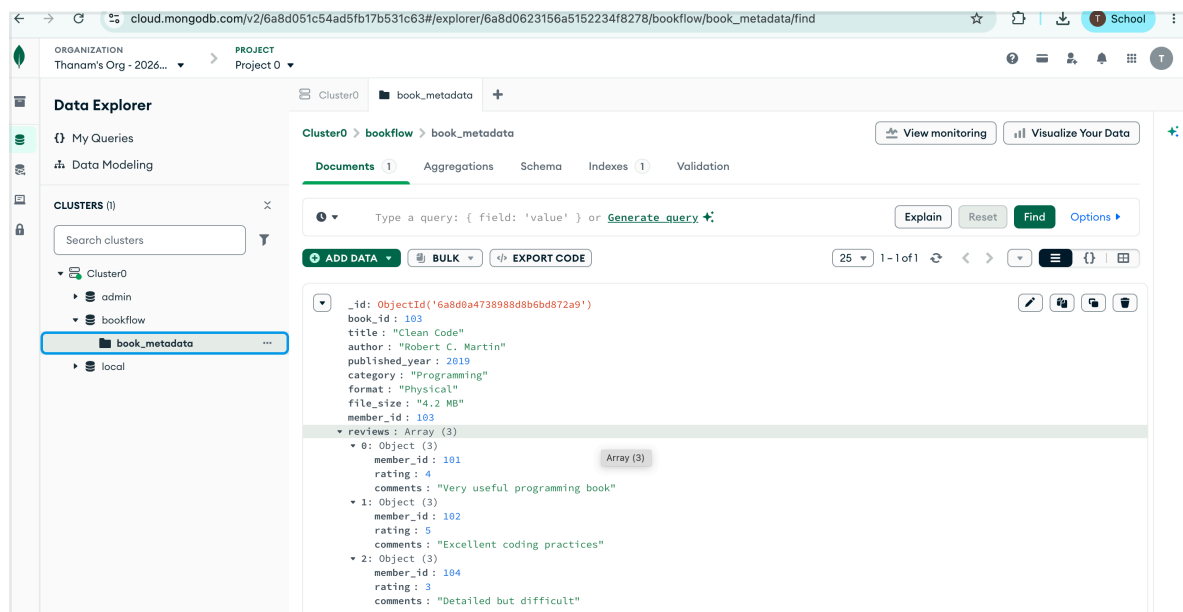
Add Data

Then select:

Insert Document

Step 3 — Delete the existing sample JSON

```
{
  "book_id": 103,
  "title": "Clean Code",
  "author": "Robert C. Martin",
  "published_year": 2019,
  "category": "Programming",
  "format": "Physical",
  "file_size": "4.2 MB",
  "member_id": 103,
  "reviews": [
    {
      "member_id": 101,
      "rating": 4,
      "comments": "Very useful programming book"
    },
    {
      "member_id": 102,
      "rating": 5,
      "comments": "Excellent coding practices"
    },
    {
      "member_id": 104,
      "rating": 3,
      "comments": "Detailed but difficult"
    }
  ]
}
```

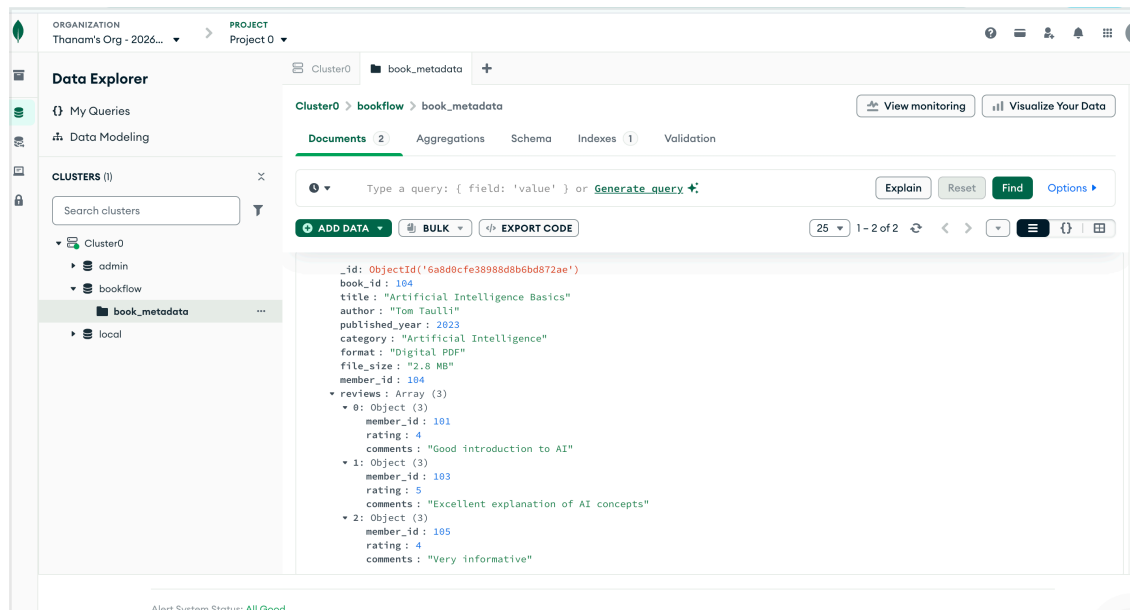


Document 4: Artificial Intelligence Basics

Click:

Add Data → Insert Document

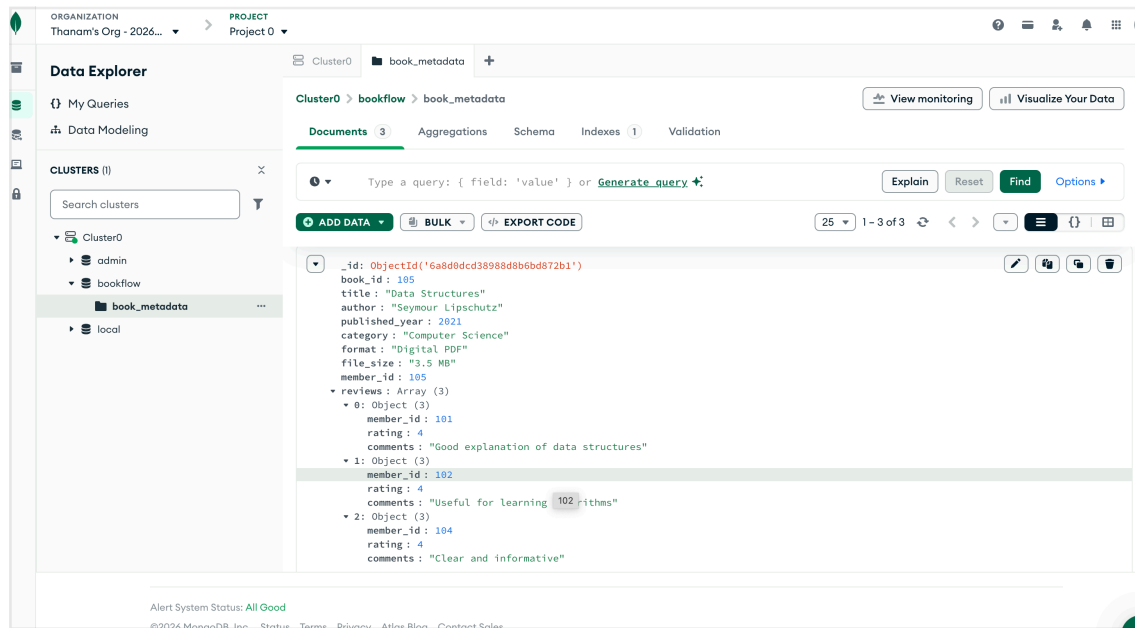
```
{
  "book_id": 104,
  "title": "Artificial Intelligence Basics",
  "author": "Tom Taulli",
  "published_year": 2023,
  "category": "Artificial Intelligence",
  "format": "Digital PDF",
  "file_size": "2.8 MB",
  "member_id": 104,
  "reviews": [
    {
      "member_id": 101,
      "rating": 4,
      "comments": "Good introduction to AI"
    },
    {
      "member_id": 103,
      "rating": 5,
      "comments": "Excellent explanation of AI concepts"
    },
    {
      "member_id": 105,
      "rating": 4,
      "comments": "Very informative"
    }
  ]
}
```



Document 5: Data Structures

1. Click **Add Data**
2. Click **Insert Document**
3. Paste:

```
{
  "book_id": 105,
  "title": "Data Structures",
  "author": "Seymour Lipschutz",
  "published_year": 2021,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "3.5 MB",
  "member_id": 105,
  "reviews": [
    {
      "member_id": 101,
      "rating": 4,
      "comments": "Good explanation of data structures"
    },
    {
      "member_id": 102,
      "rating": 4,
      "comments": "Useful for learning algorithms"
    },
    {
      "member_id": 104,
      "rating": 4,
      "comments": "Clear and informative"
    }
  ]
}
```



Step 6 — Check Documents

1. Stay inside **bookflow** → **book_metadata**.
2. Click the **Documents** tab.
3. Check that you can see:
 - The Martian
 - Dune
 - Clean Code
 - Artificial Intelligence Basics
 - Data Structures

You should see **Documents: 5**.

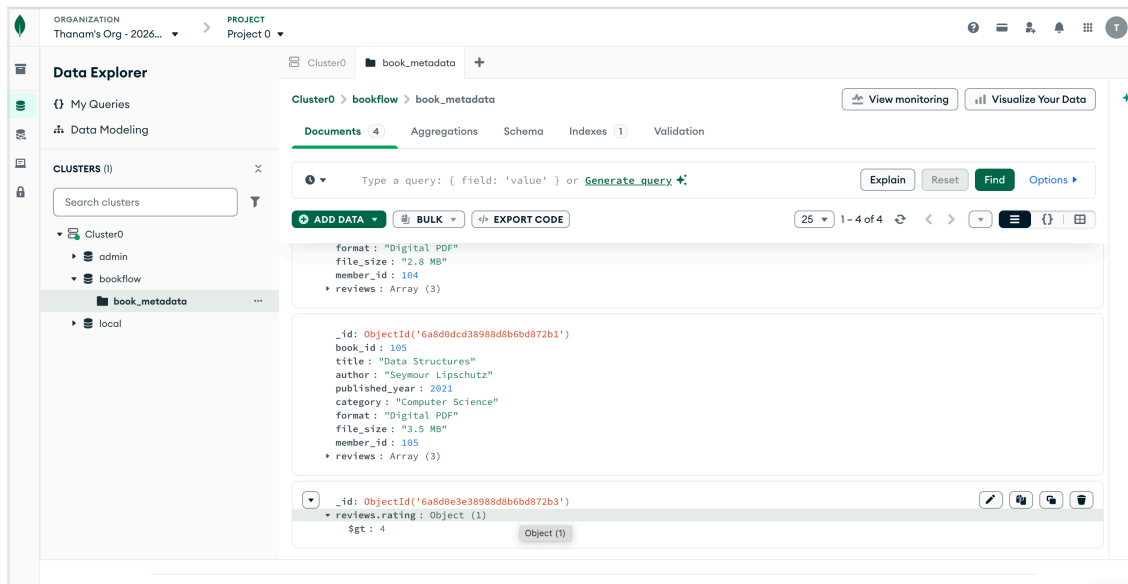
Step 7 — \$gt Query

Now in the **Filter** box at the top, enter:

```

{
  "reviews.rating": {
    "$gt": 4
  }
}

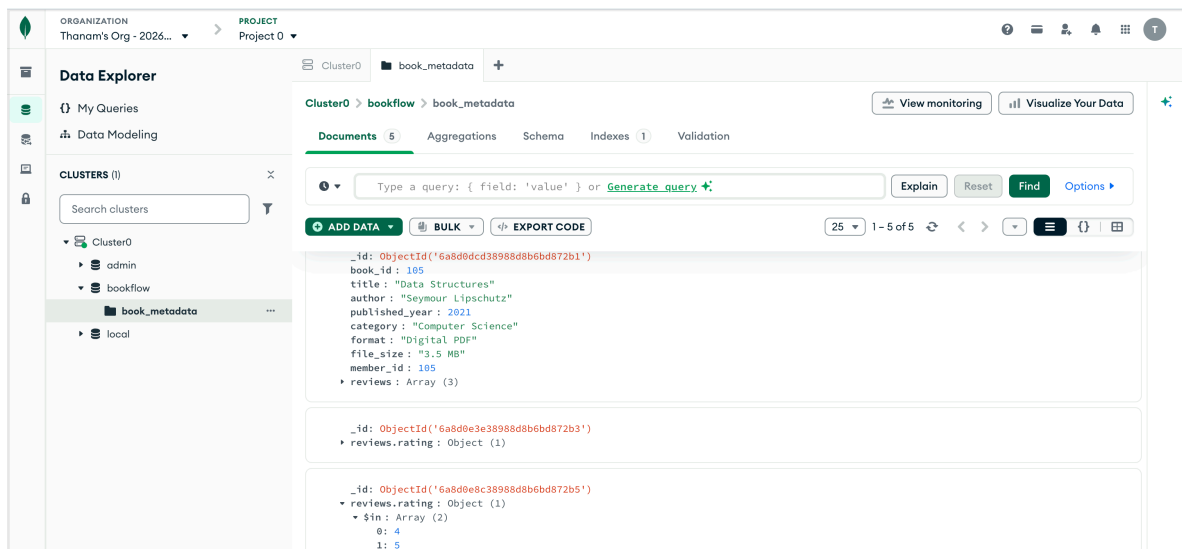
```



Step 8 — \$in Query

In the **Filter** box, replace the previous query with:

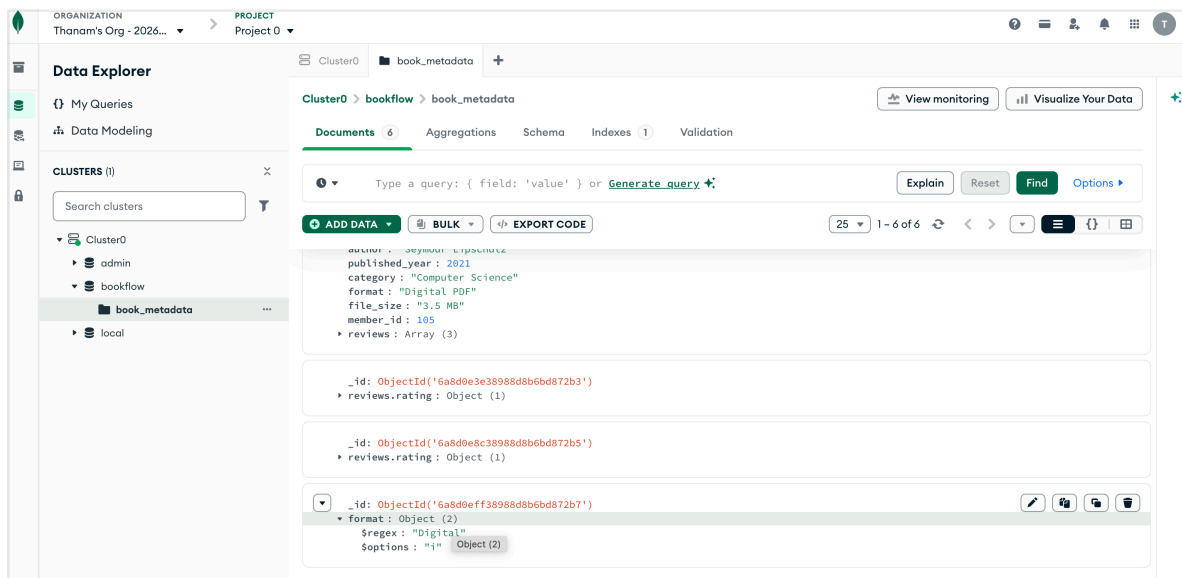
```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```



Step 9 — Regex Query

In the **Filter** box, remove the previous query and enter:

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```



Step 10 — Aggregation

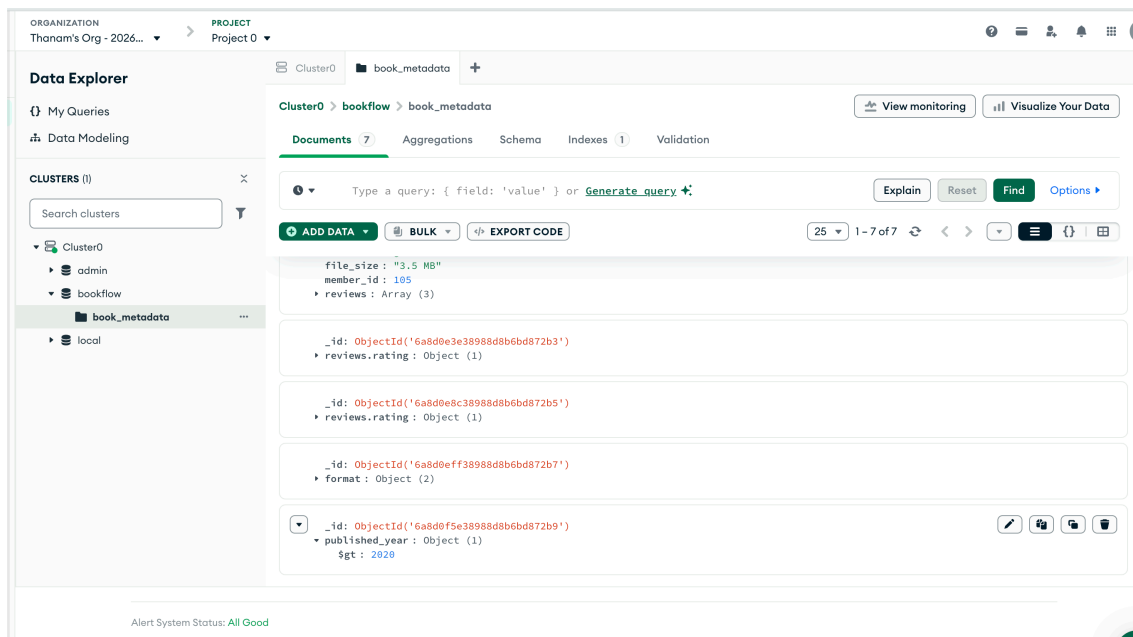
Now click the **Aggregations** tab.

We will add **4 stages**.

Stage 1 — \$match

1. Click **Add Stage**.
2. Select **\$match**.
3. Enter:

```
{
  "published_year": {
    "$gt": 2020
  }
}
```

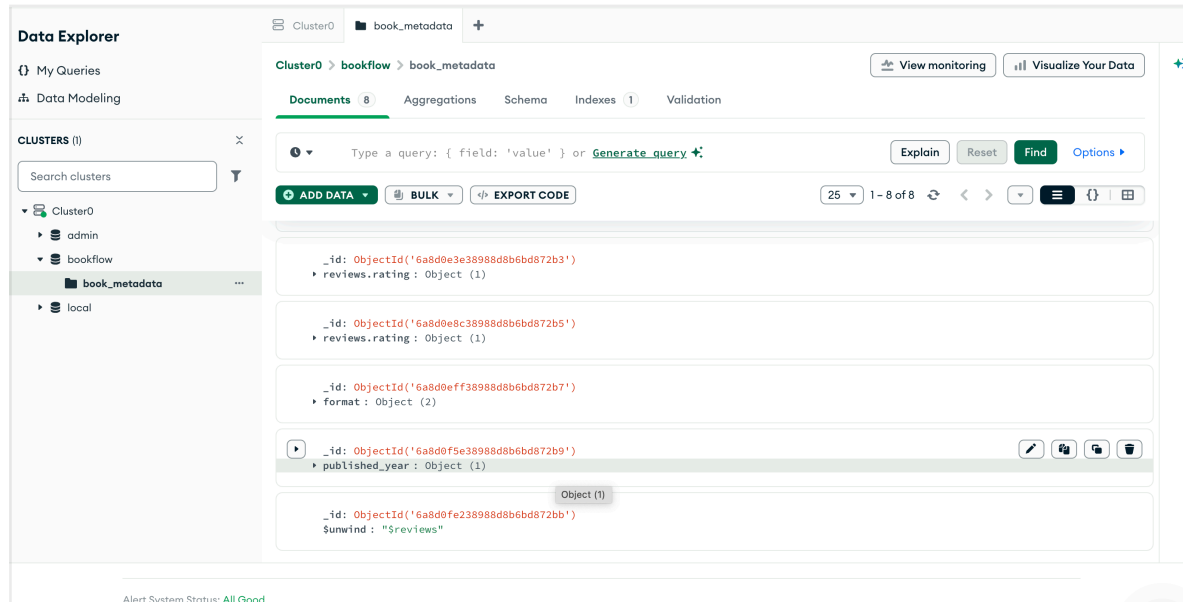


Step 11 — \$unwind

Now continue in **Aggregations**.

1. Click **Add Stage**.
2. Select **\$unwind**.
3. Enter:

\$reviews

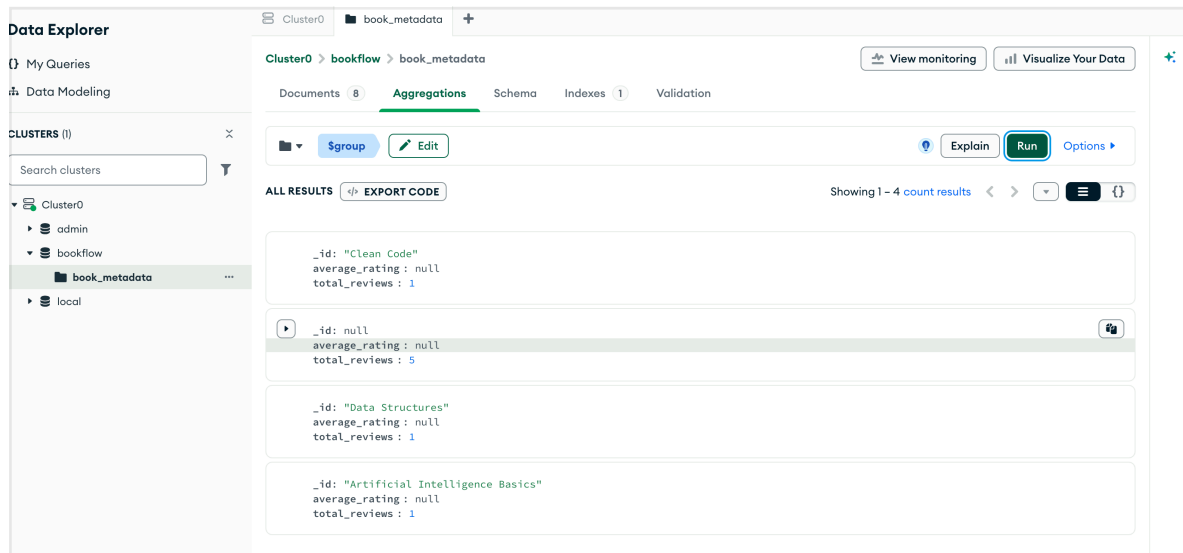


Step 12 — \$group

In the **Aggregations** tab:

1. Click **Add Stage**
2. Select **\$group**
3. Enter:

```
{
  "_id": "$title",
  "average_rating": {
    "$avg": "$reviews.rating"
  },
  "total_reviews": {
    "$sum": 1
  }
}
```

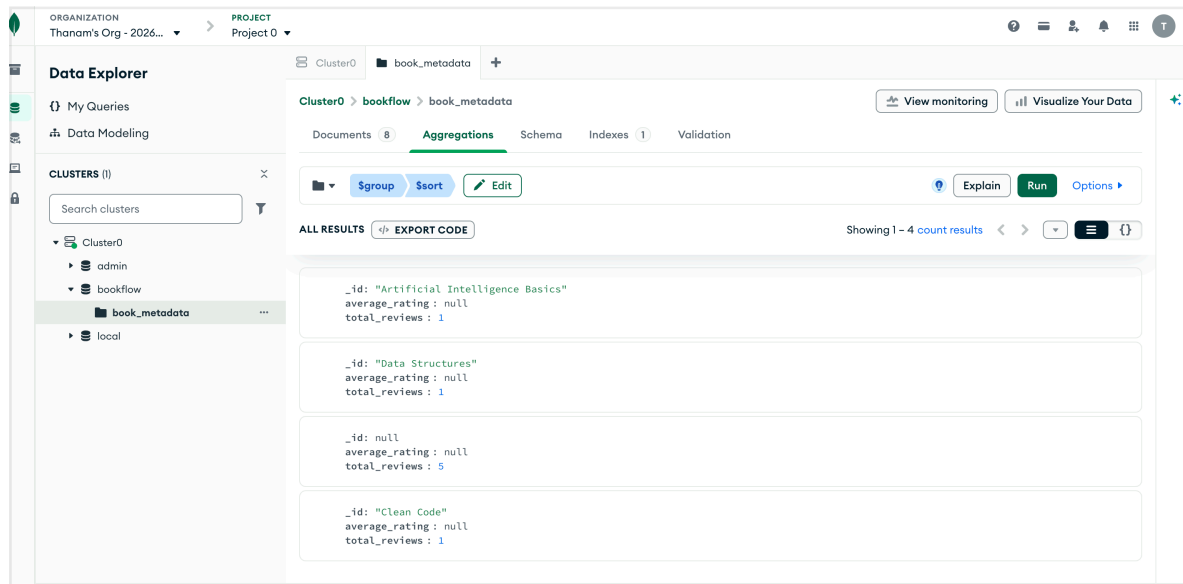


Step 13 — \$sort

Now add the final aggregation stage.

1. Click **Add Stage**.
2. Select **\$sort**.
3. Delete the default template.
4. Paste:

```
{
  "average_rating": -1
}
```



Step 14:

```
db.book_metadata.find({
  "$text": {
    "$search": "excellent"
  }
})
```

Step 15 — Text Search

Since you created the **Text Index**, now run the search.

If you are in **Mongosh**, enter:

```
db.book_metadata.find({
  $text: {
    $search: "excellent"
  }
})
```

Step 16 — explain()

Now run this in **Mongosh**:

```
db.book_metadata.find({
  $text: {
    $search: "excellent"
  }
}).explain("executionStats")
```

Step 17 — Check the result

In the explain() output, find:

IXSCAN

Meaning

- **IXSCAN** → MongoDB used the index
- **COLLSCAN** → MongoDB scanned the whole collection

Result

The Week-3 MongoDB practical was successfully completed.

- **Database:** bookflow
- **Collection:** book_metadata
- **Documents inserted:** 5
- **Queries performed:** \$gt, \$in, Regex
- **Aggregation:** \$match → \$unwind → \$group → \$sort
- **Text index:** Created on reviews.comments
- **Text search:** "excellent"
- **Index analysis:** explain("executionStats")
- **Expected index stage:** IXSCAN

Conclusion: MongoDB successfully performed document queries, aggregation, text search, and index-based query analysis.